Generation Date: 01-Dec-2024 05:06:51 (applicant copy)

THE UNIVERSITY OF HONG KONG

Application for Admission to the Doctor of Philosophy (PhD) and Master of Philosophy (MPhil) Programmes

Round of Admission: Main Round for 2025–26 Admission
(Application submitted on: 30/11/2024; username: zhouxiangru)

1. Programme Apply For

Programme:	PhD
Study mode:	Full-time
Proposed faculty /Interdisciplinary area:	Li Ka Shing Faculty of Medicine
Proposed department/area of study:	School of Biomedical Sciences
Proposed discipline:	
Proposed field of study:	Brain Computer Interface
Proposed supervisor(s):	
Proposed registration date: (DD/MM/YYYY)	1/9/2025
Apply for joint PhD programme:	Yes
Partner institution of joint PhD programme:	King's College London
Apply for work-study programme:	No
Partner Institute of work-study programme:	
Apply for PGS:	Yes
Apply for HKPFS:	No
HKPFS reference number:	

2. Personal Information

Title:	Mr.
Surname:	Zhou
Given name:	Xiangru
Middle name:	
Full name in Chinese:	周相如
Sex:	M
Date of birth (DD/MM/YYYY)	31/7/1990
HKID card number:	F601647(9)
Passport number:	
Chinese National ID:	450881199007318019
HKU student number:	
Country of residence:	China
Nationality (Place of origin):	China
Correspondence address:	Room 2205, Building 12, Qinghaian Garden,
	Nansha Street, Nansha District,
	Guangzhou, Guangdong, China.

Postal code:	511458	
Email address:	**(1)** xiangru.zhou@connect.polyu.hk	
	(2) JohnsonZhouXR@outlook.com	
Contact phone number:	**(1)** +8618502091962	
	(2) 85264820616	
Right of abode:	N	
Need visa/entry permit:	Y	
Type of visa/entry permit:	Student visa/entry permit	
First issued date of visa/entry permit (DD/MM/YYYY):		

3. Academic Qualification

3.1 Current Academic Studies

[Only one record is allowed. Applicants with more than one current academic studies are advised to provide the complete information in Section 7 Additional Information.]

(1)	Level of qualification:	Other Master's degree (Full-time)
	Awarding institution:	Hong Kong Polytechnic University (The)
	Awarding country:	Hong Kong(SAR)of PRC
	Major subject:	Microelectronics Technology & Materials
	Start date of attendance:	9/2024
	Year(s) of study completed:	1
	Expected date of award (DD/MM/YYYY):	1/7/2025
	Cumulative grade point average (CGPA):	
	Average mark:	
	Language of instruction:	English
	Qualification obtained by:	By Coursework

3.2 Previous Academic Studies

(1)	Level of qualification:	Bachelor's Degree (Full-time)
	Awarding institution:	Xi'an University of Technology
	Awarding country:	China
	Major subject:	Microelectronics
	Date of attendance:	From 9/2009 to 7/2013
	Year(s) of study completed:	4
	Status:	Graduated
	Title of award:	Bachelor of Engineering
	Date of award (DD/MM/YYYY):	1/7/2013
	Honours classification:	Not Applicable
	Cumulative grade point average (CGPA):	
	Average mark:	
	Language of instruction:	Not English
	Qualification obtained by:	By Coursework

4. Professional Qualification, Prize/Award and Work Experience

4.1 Professional Qualification, Prize and Award
Nil

4.2 Work Experience

	Name of organisation	Position	Mode	Employment period (DD/MM /YYYY)	Duration of employment (in days)	Major responsibilities
(1)	SmartMore Corporation Limited	Software Engineer	Full-time	21/5/2020 to 1/9/2024	1565	Software Engineer, C++ & python coder and total solution provider. Leader of a team of four.

5. English Language and Other Test Results
5.1 English Language Test

English language test	Date of test (MM/YYYY)	Overall score/grade	Appointment number
TOEFL:			
IELTS (Academic module):	3/2024	6.5	-
GCE:			-
IGCSE:			-
CPE:			-

5.2 Other Tests
Nil

6. Referees

(1)	Full name:	Prof. Jun YIN
	Position:	Associate Professor
	Organisation and department:	The Hong Kong Polytechnic University
	Personal webpage:	
	Relationship to applicant:	Lecturer
	Email:	jun.yin@polyu.edu.hk
	Contact phone number:	852 2766 5684
	Postal address:	CD604
		11 Yuk Choi Road, Hung Hom,
		KLN, Hong Kong.

(2)	Full name:	Dr. Ruiyu LI
	Position:	Department supervisor
	Organisation and department:	SmartMore Corporation Limited
	Personal webpage:	
	Relationship to applicant:	Former supervisor
	Email:	ryli@smartmore.com
	Contact phone number:	86 13058014285

	Postal address:	22F, Building T2, Qianhai Kerry Center,
		Nanshan street, Nanshan District,
		Shenzhen City, Guangdong Province, China.

7. Additional information

Nil

学士学位证书

周相如　，男，1990 年 7 月 31 日生。在　西安理工大学

微电子学　　　　　　　　专业完成了本科学习计划，业已

毕业，经审核符合《中华人民共和国学位条例》的规定，授予 工学

学士学位。

西安理工大学

校　　长

学位评定委员会主席

证书编号：1070042013001855

二〇一三 年 七 月 三 日

（普通高等教育本科毕业生）

西安理工大学 学生学位证明

XI'AN UNIVERSITY OF TECHNOLOGY

姓名	周相如
性别	男
出生日期	1990-07-31
入学时间	2009-09-10
毕业时间	2013-07-03
系别	自动化与信息工程学院
专业	微电子学

经审核符合《中华人民共和国学位条例》规定，授予工学学士学位。

学位委员会主席：刘丁

学校（盖章）：

西安理工大学

学位证书号:1070042013001855　　打印时间:2024-6-4

XI'AN UNIVERSITY OF TECHNOLOGY
Bachelor's Degree Certificate

Name	Zhou Xiangru
Gender	Male
Date of Birth	1990-07-31
Data of Enrollment	2009-09-10
Date of Graduation	2013-07-03
Department	Faculty of Automation and Information Engineering
Speciality	Microelectronics

In accordance with" The Regulation Concerning Academic Degrees in the People's Republic of China" , the student is awarded the Bachelor of Engineering.

Chairman of Degree's Committee	Liu Ding
Seal of University	

Degree certificate number : 1070042013001855

Date : Jun. 4, 2024

西安理工大学 XI'AN UNIVERSITY OF TECHNOLOGY 学生成绩单

学号	3090433019	姓名	周相如	性别	男	出生日期	1990-07-31	院系	自动化与信息工程学院	专业	微电子学
国籍	中国	学制	4年			入学时间	2009-09-10	毕业时间	2013-07-03	备注	否 （是/否休学或留级）

2009~2010学年

上学期

课程名称	学分	成绩
英语	4.5	86
中国近现代史纲要	1.0	62
高等数学	5.5	73
大学计算机基础	2.5	100
体育	1.0	80
思想品德修养与法律基础	2.0	61
军训	0.5	85

下学期

课程名称	学分	成绩
C语言程序设计	3.5	94
C语言程序设计课程设计	1.0	中
电子技术查新训练	1.0	优秀
英语	4.5	80
马克思主义基本原理	2.0	90
高等数学	6.0	77
线性代数	2.5	84
物理实验	1.0	良好
大学物理	3.5	84
图学基础及CAD	3.0	86
高分子世界概论	2.5	优秀
体育	1.0	85
军事理论	0.5	88
艺术导论	1.5	66

2010~2011学年

上学期

课程名称	学分	成绩
计算机软件基础	3.5	84
电路	4.0	80
电路实验	0.5	中
英语	4.5	74
日语	2.5	90
毛泽东思想和中国特色社会主义理论体系	3.0	69
复变函数与积分变换	2.5	61
概率论及数理统计	3.0	87
物理实验	1.0	优秀
大学物理	4.0	78
工程训练	3.0	良好

下学期

课程名称	学分	成绩
新型复合材料	1.5	89
模拟电子技术	4.0	82
模拟电子技术实验	0.5	中
理论物理基础	3.5	61
数理方程	2.0	87
电子线路CAD综合训练	2.0	良好
英语	5.0	80
中国传统文化	2.0	90
工程技术综合实践	2.0	优秀

2011~2012学年

上学期

课程名称	学分	成绩
数字电子技术	4.0	91
数字电子技术实验	0.5	良好
数字电子技术课程设计	1.0	优秀
固体物理	3.5	83
半导体物理	4.5	84
生产实习	2.0	良好
半导体基础实验	1.0	良好
高频电子线路	2.5	76
计算机网络与通信	2.5	84
体育	1.0	79

下学期

课程名称	学分	成绩
微机原理及应用	4.0	86
集成电路工艺原理	3.5	94
微电子技术基础	4.0	83
生产实习	2.0	优秀
微电子技术综合实践	2.0	良好
半导体材料	2.0	71
MCS51单片机原理及应用	2.0	82
光电子学原理	2.5	90
体育	1.0	90

2012~2013学年

上学期

课程名称	学分	成绩
半导体集成电路	4.0	86
集成电路设计技术	3.0	76
半导体专业实验	1.5	优秀
集成电路设计实践	2.0	良好
检测与传感技术	2.0	81
射频集成电路设计技术	2.5	86
电力半导体器件	2.0	73

下学期

课程名称	学分	成绩
毕业设计	17.0	良好

等级考试*

CET-4		466
CET-6		465

注: 0-59 分为不及格；60-64 分为及格；65-74 分为中；75-84 分为良好；85-100 分为优秀。

学校（章）：　　　　审核：　　　　制表：档案馆　　　　打印日期:2024 年6月4日

XI'AN UNIVERSITY OF TECHNOLOGY STUDENT'S ACADEMIC RECORD

Stu NO.	3090433019	**Name**	Zhou Xiangru	**Gender**	Male	**Birthday**	1990-07-31	**Department**	Faculty of Automation and Information Engineering	**Speciality**	Microelectronics
Nationality	Chinese	**Education System**		4		**Enrollment**	2009-09-10	**Graduation**	2013-07-03	**Suspension/repeater**	/

First Academic Year(2009~2010)

COURSES	CREDIT	SCORE
Fall Semester		
College English	4.5	86
Outline of China's Modern History	1.0	62
Advanced Mathematics	5.5	73
Fundamentals of Computer	2.5	100
Physical Education	1.0	80
Ideological and Moral Cultivation & Basis of Law	2.0	61
Military Training	0.5	85
Spring Semester		
C Language Programming	3.5	94
Course Design of C Language Programming	1.0	74
Electronic Technology Novelty Search Training	1.0	95
College English	4.5	80
The Basic Principles of Marxism	2.0	90
Advanced Mathematics	6.0	77
Linear Algebra	2.5	84
Experiment of College Physics	1.0	84
College Physics	3.5	84
Basis of Drawing & AutoCAD	3.0	86
Introduction to the World of Polymer	2.5	95
Physical Education	1.0	85
Military Theory	0.5	88
Introduction of Art	1.5	66

Second Academic Year(2010~2011)

COURSES	CREDIT	SCORE
Fall Semester		
Foundation of Computer Software	3.5	84
Circuits	4.0	80
Experiment of Circuits	0.5	74
College English	4.5	74
Japanese	2.5	90
Mao Zedong Thought and the Theoretical System of Socialism with Chinese Characteristics	3.0	69
Functions of Complex Variables and Integral Transformation	2.5	61
Probability Theory and Mathematical Statistics	3.0	87
Experiment of College Physics	1.0	95
College Physics	4.0	78
Engineering Training	3.0	84
Spring Semester		
Advanced Composite Materials	1.5	89
Analog Electronics Technology	4.0	82
Experiment of Analog Electronic Technology	0.5	74
Foundation of Theoretical Physics	3.5	61
Equations of Mathematical Physics	2.0	87
Comprehensive Training of Electronic Circuits CAD	2.0	84
College English	5.0	80
Chinese Traditional Culture	2.0	90
Engineering Technology Comprehensive Practice	2.0	95

Third Academic Year(2011~2012)

COURSES	CREDIT	SCORE
Fall Semester		
Digital Electrical Technology	4.0	91
Experiment of Digital Electrical Technology	0.5	84
Course Design of Digital Electronic Technology	1.0	95
Solid-state physics	3.5	83
Semiconductor Physics	4.5	84
Production Practice	2.0	84
Basis Experiment of Semiconductor	1.0	84
High Frequency Electronic Circuits	2.5	76
Computer Networks and Communication	2.5	84
Physical Education	1.0	79
Spring Semester		
Microcomputer Principle and Application	4.0	86
Fundamentals of Integrated Circuit Process	3.5	94
The Foundation of Microelectronic Technology	4.0	83
Production Practice	2.0	95
Microelectronic Technology Practice	2.0	84
Semiconductor Materials	2.0	71
Principles and Applications of MCS 51 series Single Chip Micro-Computer	2.0	82
Theory of Optoelectronics	2.5	90
Physical Education	1.0	90

Fourth Academic Year(2012~2013)

COURSES	CREDIT	SCORE
Fall Semester		
Semiconductor Integrated Circuit	4.0	86
Design of Integrated Circuit	3.0	76
Professional Experiments of Semiconductor	1.5	95
Design Practice of Integrated Circuit	2.0	84
Measure and Sensor Technology	2.0	81
The Design of Radio-Frequency Integrated circuits	2.5	86
Power Semiconductor Devices	2.0	73
Spring Semester		
Graduation Design	17.0	84
Rank Examination*		
CET-4		466
CET-6		465

PS: Fail 0—59； Pass（PS）60—64； Credit （CR）65—74； Distinction （DI）75—84； High Distinction （HD）85—100

Seal of University：XI'AN UNIVERSITY OF TECHNOLOGY

Checker：

Abulation：Archives

Date：Jun. 4, 2024

西安理工大学 XI'AN UNIVERSITY OF TECHNOLOGY 学生学业成绩表

学院：自动化与信息工程学院　　专业：微电子学　　班级：微电091

姓名：周相如　　学号：3090433019

课程名称	课程性质	学分	成绩	绩点
2009-2010学年 第1学期				
英语A(2)	必修课	4.5	86	4.1
中国近现代史纲要	必修课	1.0	62	0
高等数学A(上)	必修课	6.0	73	2.8
大学计算机基础A	校级选修课	2.5	100	5
体育(1)	必修课	1.0	80	3.5
思想道德修养与法律基础	必修课	2.0	61	1.6
军训	必修课	0.5	85	4
2009-2010学年 第2学期				
C语言程序设计	必修课	3.5	94	4.9
C语言程序设计课程设计	必修课	1.0	中	2.5
电子技术查新训练	必修课	1.0	优	4.5
英语A(3)	必修课	4.5	80	3.5
马克思主义基本原理	必修课	2.0	90	4.5
高等数学A(下)	必修课	6.0	77	3.2
线性代数	必修课	2.5	84	3.9
物理实验(上)	必修课	1.0	良	3.5
大学物理A（上）	必修课	3.5	84	3.9
图学基础及CAD	必修课	3.0	86	4.1
高分子世界概论	校级选修课	2.5	优	4.5
体育(2)	必修课	1.0	85	4
军事理论	必修课	0.5	88	4.3
艺术导论	校级选修课	1.5	66	2.1
2010-2011学年 第1学期				
计算机软件基础	必修课	3.5	84	3.9
电路（1）	必修课	4.0	80	3.5
电路（1）实验	必修课	0.5	中	2.5
英语A(4)	必修课	4.5	74	2.9
日语（中级）	校级选修课	2.5	90	4.5
毛泽东思想和中国特色社会主义理论体系概论	必修课	3.0	69	2.4
复变函数与积分变换	必修课	2.5	61	1.6
概率论及数理统计B	必修课	3.0	87	4.2
物理实验(下)	必修课	1.0	优	4.5
大学物理A（下）	必修课	4.0	78	3.3
工程训练 A	必修课	3.0	良	3.5
2010-2011学年 第2学期				
新型复合材料	校级选修课	1.5	89	4.4
模拟电子技术	必修课	4.0	82	3.7
模拟电子技术实验	必修课	0.5	中	2.5
理论物理基础	必修课	3.5	61	1.6
数理方程	必修课	2.0	87	4.2
电子线路CAD综合训练	必修课	2.0	良	3.5
中国传统文化	校级选修课	2.0	90	4.5
英语A(5)	必修课	4.5	80	3.5
工程技术综合实践	校级选修课	2.0	优	4.5
2011-2012学年 第1学期				
数字电子技术	必修课	4.0	91	4.6
数字电子技术实验	必修课	0.5	良	3.5
数字电子技术课程设计	必修课	1.0	优	4.5
固体物理	必修课	3.5	83	3.8
半导体物理	必修课	4.5	84	3.9
生产实习1(微电)	必修课	2.0	良	3.5
半导体基础实验	必修课	1.0	良	3.5
高频电子线路D	必修课	2.5	76	3.1
计算机网络与通信B	院级选修课	2.5	84	3.9
体育(3)	必修课	1.0	79	3.4
2011-2012学年 第2学期				
微机原理及应用A	院级选修课	4.0	86	4.1
集成电路工艺原理	必修课	3.5	94	4.9
微电子技术基础	必修课	4.0	83	3.8
生产实习2(微电)	必修课	2.0	优	4.5
微电子技术综合实践	必修课	2.0	良	3.5
半导体材料	院级选修课	2.0	71	2.6
MCS51单片机原理及应用	院级选修课	2.0	82	3.7
光电子学原理	院级选修课	2.5	90	4.5
体育(4)	必修课	1.0	90	4.5
2012-2013学年 第1学期				
半导体集成电路A	必修课	4.0	86	4.1
集成电路设计技术A	必修课	3.0	76	3.1
半导体专业实验	必修课	1.5	优	4.5
集成电路设计实践	必修课	2.0	良	3.5
检测与传感技术(电子、微电	院级选修课	2.0	81	3.6
射频集成电路设计技术	院级选修课	2.5	86	4.1
电力半导体器件	院级选修课	2.0	73	2.8
2012-2013学年 第2学期				
毕业设计(微电)	必修课	17.0	良	3.5

以下空白

已获得总学分:184.0　必修课:150.0　校级选修课:14.5　院级选修课:19.5

平均学分绩点:3.64　加权平均成绩:81.18　CE[illegible]:466;

审核：张冠华　　院系(盖章)：自动化与信息工程学院　　教务处盖章：西安理工大学 学生学习成绩专用章　　打印日期：2024-05-30

Advancing Brain-Computer Interface (BCI) Technology

Background and Motivation

I am currently a Master's student in Microelectronics Technology and Materials at The Hong Kong Polytechnic University. Before pursuing my Master's degree, I accumulated over ten years of professional experience in developing image processing algorithms and AI-based image recognition solutions. I previously worked as an Algorithm Engineer and R&D Lead at notable organizations, including the publicly listed company OPT and the prominent AI firm SmartMore Technology.

My skill set includes programming, engineering debugging, setting up and deploying integrated development environments, data collection and preprocessing, parameter optimization, model training and testing, result analysis and refinement, as well as replicating and enhancing cutting-edge technologies. I have successfully filed and been granted more than ten invention patents. Additionally, I possess expertise in designing Bluetooth communication devices, drawing circuit schematics, and creating 3D models and printed components. My prior projects include using AI technology to control multi-axis robotic arms and other complex systems.

I have long been interested in Brain-Computer Interface (BCI) technology, which I regard as a guiding vision for my career development. To me, BCI is a multidisciplinary field that requires a balanced focus: 50% on engineering and computer science (including AI and electronics), 20% on materials science and microphysics, and 30% on bioengineering and medical science (especially neuroscience and brain science). Over the past decade, I have concentrated on software development, image processing, and AI applications while maintaining a strong passion for computer and electronic technologies. With the rapid advancements in BCI, I believe now is the perfect time to take a decisive step forward into this field.

Current Focus and Future Plans

To transition into BCI research, I have revisited foundational technologies, such as cochlear implants and OpenBCI systems, while reorganizing my technical expertise. With a solid background in image processing, software development (mainly in C++ and Python), and AI applications, I decided to pursue a Master's degree in Semiconductor Materials and Technology to address my knowledge gaps in nanomaterials and microphysics. These interdisciplinary skills are critical for improving the precision and accuracy of signal acquisition in BCI, where electrode materials and their interactions with biological signals are particularly important.

Looking ahead, I plan to complement my engineering expertise with foundational knowledge in biology, neuroscience, and medical science, enabling me to collaborate more effectively with experts in BCI teams. My current research focus will continue to center on software development, signal acquisition and processing, and model training and optimization. By analyzing large datasets of neural signals, I aim to identify patterns and use AI techniques to enhance signal recognition accuracy. Moreover, I am particularly interested in exploring how brain signals can control robotic arms, combining my prior experience to develop better rehabilitation solutions for patients.

Research Goals

If given the opportunity to pursue a PhD, I aim to achieve the following research objectives:

- **Data Acquisition and Analysis**: Collect and preprocess large volumes of neural signal data, ensuring its quality through meticulous cleaning, classification, and validation.
- **Model Development**: Design and refine brain signal recognition models, leveraging AI technology to improve accuracy, robustness, and stability.
- **Real-World Applications**: Investigate the feasibility of brain-controlled devices, particularly in integrating BCI technology with robotics.
- **Interdisciplinary Collaboration**: Combine software engineering with neuroscience, brain science, and materials science to deliver innovative BCI solutions, especially focusing on advancements in electrode technology.

Through these efforts, I aspire to contribute to the development and clinical validation of BCI technologies, ultimately making a meaningful societal impact.

Challenges and Preparedness

I fully recognize the challenges in this field, such as low initial recognition rates for models, stability issues, and difficulties in transferring results across devices or environments. However, my extensive professional experience equips me well to address these challenges. I understand the critical role of data quality in determining the success of a model and how improvements in sensor precision, signal processing techniques, and experimental setups can significantly enhance research outcomes.

Furthermore, I am aware of the importance of interdisciplinary collaboration. For instance, in BCI, electrode materials play a decisive role in signal acquisition accuracy and reliability, and understanding their properties and interactions with biological systems is crucial. Similarly, collaborating with medical professionals requires a solid grasp of medical protocols and effective communication. By equipping myself with knowledge in neuroscience, materials science, and engineering, I am preparing methodically to overcome these challenges.

Commitment to BCI and the Greater Bay Area

I have deep ties to the Greater Bay Area, with my family based in Nansha District, Guangzhou, and eight years of professional experience in Shenzhen, including four years as an entrepreneur. I am fluent in Cantonese, Mandarin, and English (with basic proficiency in Japanese), which enables me to thrive in the region's international innovation ecosystem.

The Greater Bay Area provides a unique environment for advancing BCI technology, offering world-class medical facilities, abundant data resources, and a vibrant culture of innovation. I am committed to establishing myself in this region, channeling my research findings back into the local community to drive technological development and enhance social welfare.

Conclusion

Thank you for taking the time to review my research proposal. I am eager to contribute to the advancement of BCI technology and look forward to the opportunity to collaborate with like-minded researchers. Please feel free to contact me if you have any questions or need additional information.

IELTS™

Test Report Form

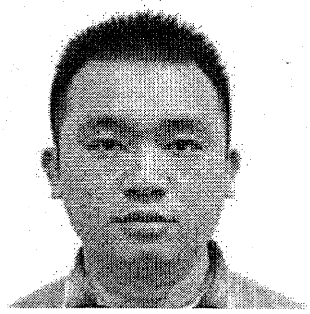
ACADEMIC

NOTE *Admission to undergraduate and post graduate courses should be based on the ACADEMIC Reading and Writing Modules. GENERAL TRAINING Reading and Writing Modules are **not** designed to test the full range of language skills required for academic purposes. It is recommended that the candidate's language ability as indicated in this Test Report Form be re-assessed **after two years** from the date of the test.*

Centre Number	CN002	Date	10/MAR/2024	Candidate Number	137676

Candidate Details

Family Name	ZHOU
First Name	XIANGRU
Candidate ID	450881199007318019



Date of Birth	31/07/1990	Sex (M/F)	M	Scheme Code	Private Candidate

Country or Region of Origin	CHINA (PEOPLE'S REPUBLIC OF)
Country of Nationality	
First Language	CHINESE

Test Results

Listening	Reading	Writing	Speaking	Overall Band Score	CEFR Level
6.0	6.5	6.0	7.0	6.5	B2

Administrator Comments

idp Education Pty Ltd

Centre stamp

BRITISH COUNCIL
EXAMINATIONS SERVICES
CHINA

Validation stamp

VALIDATION STAMP IELTS CAMBRIDGE ENGLISH/BRITISH COUNCIL/IDP:IA

Administrator's Signature

Date 13/03/2024

Test Report Form Number 23CN137676ZHOX002A

BRITISH COUNCIL idp Cambridge Assessment English

Certificate of Admission

The Hong Kong Polytechnic University

香港理工大學

This is to certify that

ZHOU Xiangru

has been admitted to study a master's degree programme in Microelectronics Technology and Materials in the academic year 2024/25

Jin-Guang Teng
President

August 2024

To learn and to apply, for the benefit of mankind 開物成務 勵學利民

Xiangru Zhou

Email: xiangru.zhou@connect.polyu.hk | Telephone: (+86) 18502091962, (852) 64820616

EDUCATION

The Hong Kong Polytechnic University — **Hong Kong**
MSc Microelectronics Technology & Materials — *Sep. 2024 – Present*

Xi'an University of Technology — **Xi'an, China**
Bachelor of Microelectronics — *Sep. 2009 – Jun. 2012*

- **GPA: 3.64/5.0**
- **Average Score:** 81.18/100
- **Honours**: 2nd Prize 8th Xi'an High-tech "Challenge Cup" Shaanxi Province University Student Extracurricular Academic Science and Technology Works Competition; 1st and 3rd Prize University 19th Extracurricular Academic Science and Technology Works Competition; Special Prize & 2nd Prize University 19th "LiAo Cup"; 1st Prize University 18th "LiAo Cup"; University Innovation Achievement 2nd Prize, University 3rd Prize Scholarship

PROFESSIONAL EXPERIENCE

SMARTMORE CORPORATION LIMITED — **Shenzhen, China**
Software Engineer — *May 2020 – Sep. 2024*

AI-Driven Robotic Arm Project
Leader — *May 2024 – Sep. 2024*

- Branch 1: Traditional Hand-Eye Calibration + Image Recognition
  - Implemented 3D scenarios: arbitrary object recognition and grasping, sketching based on verbal commands, and interactive stacking toys with users.
  - Integrated an Intel Realsense camera, a structured light 3D camera, a 7-axis robotic arm with a gripper, two servers for large-model deployment, and a PC for control and ASR-based voice input.
  - Deployed conversational and vision models to enable seamless interaction between visual, auditory, and motor systems.
  - Achieved high precision in object localization and vision-guided robotic operations.
- Branch 2: Transformer-Based Action Prediction
  - Utilized a Transformer model to predict robotic arm actions by joint data, CNN features, and task descriptions.
  - Enabled real-time responses without hand-eye calibration, adapting to dynamic environments.
  - Tackled computational challenges and dataset requirements for effective training and deployment.

Wafer ID Reader Wafer OCR Code Reader Integrated Machine Project
Leader — *Oct. 2021 – Sep. 2024*

- The first deep learning-based OCR wafer character recognition tool in the semiconductor industry. Convenient use and no need to adjust the recognition parameters compare to its competing products.
- Collaborated with the algorithm team to devise strategies, conduct experiments, and select optimal algorithmic solutions.
- Modified software and SDK interfaces and functionalities based on the product manager's research and user requirements.
- Implemented end-to-end closed-loop process from requirement analysis to deployment, including evaluation, experimentation, algorithm/software/SDK development, testing, and deployment.
- Same recognition rate level as that of industry leader Cognex products in normal cases, partially exceeds its recognition rate (100% versus 99.5%) in the case of fixed format recognition.

Defect Detection System for SONY Labels
Second-hand Collaborator — *Feb. 2023 – May 2023*

- Conducted Proof of Concept (POC) using client samples to validate system performance.
- By combining the template image and the image to be checked together to form a two-channel image, defects are marked and trained on this basis, the accuracy has been improved by 2%.

BGI Genomics Reagent Bottle Testing
Collaborator — *Oct. 2022 – Feb. 2023*

- Designed algorithmic solutions and implemented the AI model into software for real-time monitoring of reagent bottle caps, ensuring seal integrity and smoothness during the manufacturing process.
- Conducted Optical Character Recognition (OCR) for character detection on reagent bottle bodies, effectively identifying characters on various colored backgrounds of medication and detecting printing defects.

Specific String OCR System for Arbitrary Orientations On Tags
Designer — *Jun. 2021 – Feb. 2022*

- Designed the composition of models responsible for the task, leveraging the existing MobileNet training framework in PyTorch.Trained four models individually: label localization, orientation determination, character localization, and recognition.
- Developed a software development kit (SDK) in Visual Studio to facilitate integration and usage of the OCR system.
- Designed and implemented a user-friendly interface using the QT framework to enhance the usability and accessibility of the OCR system.

Character Recognition System for Apple Watch Bands Using Laser Engraving Technology
Designer — *May. 2020 – May 2021*

• Trained character localization and classification models on a four-card 2080Ti server running Ubuntu, utilizing the MobileNet architecture.
• Employed the ONNX Runtime (ORT) for model deployment, integrating a C++ SDK for inference on the server and developing a user-friendly software interface.
• Collaborated with colleagues to align and optimize the SDK, including pre-processing and post-processing workflows.
• Precision and recall were above 99.7% respectively.

Defect Detection System for Apple Watch Bezels Using 3D Imaging Technology

Designer *Jun. 2020 – Jun. 2020*

• Implemented the conversion of 3D point cloud data into 2D heatmaps for analysis purposes.
• Designed and programmed the user interface for the software application.

SHENZHEN PHDI CORPORATION LIMITED **Shenzhen, China**

Co-founder, CTO *Jan. 2017 - May 2020*

Android Software for A Car-Mounted Mobile Phone Controller

Designer *Jan. 2017 – May 2020*

• Managed PCB production designed Bluetooth chip-related programs and conducted 3D design and printing for the product's appearance.
• Enabled rapid switching between navigation, music, and WeChat applications in a car environment.
• Implemented automatic sending and receiving of WeChat messages.
• Integrated features for one-touch activation of voice navigation and song search.
• Designed and developed a comprehensive product including a mobile application and a Bluetooth controller.

Face Recognition and Tracking Software on The Android Platform

Designer *Jan. 2017 – May 2017*

• Utilized OpenCV, Caffe, Idlib, and VGG net deep learning neural network for face localization and gender recognition.
• Integrated BP neural network for gender recognition within the face detection algorithm.
• Applied the developed algorithm in systems for estimating the attention of different gender groups towards billboards and for access control systems, resulting in the acquisition of a patent.
• Adapted the algorithm for integration with security cameras to track familiar/strange faces indoors, aiding in the detection of potential security threats.
• Successfully delivered the developed solution to a renowned domestic security camera company.

OPT MACHINE VISION TECH CO., LTD. **Guangdong, China**

Image processing algorithm engineer *Apr. 2014 – Jan. 2017*

One-Dimensional Barcode Localization and Recognition Algorithm

Designer *Apr. 2014 – Jan. 2017*

• Achieved automatic localization of barcodes in approximately 10ms on a PC equipped with an i3 processor, capable of identifying barcodes in complex images of up to 2 million pixels.

Two-Dimensional Barcode Localization and Recognition Algorithm

Designer *Jan. 2015 – Jan. 2017*

• Implemented functionality to locate and recognize QRCode, DataMatrix, and other two-dimensional barcodes.
• Utilized Google's open-source libraries ZXing/Zbar and made necessary modifications for enhanced performance

Rapid Edge And Circle Detection Algorithm

Designer *Jan. 2015 – Jan. 2017*

• Utilized non-Hough fitting method to swiftly fit specified parameters of lines and circles in designated directions, implementing the algorithm to effectively recognize lines and circles in blurry edge images of up to 2 million pixels.
• Achieved real-time identification within 10ms on a PC with an i3 processor and 8GB of memory.

ADDITIONAL INFORMATION

• **Language**: English (Fluent; IELTS: 6.5), Japanese (N5)
• **Programming Language**: C, C++, Python, Java, Java Script, HTML, CSS, PHP
• **Development tools:** QT, Visual Studio
• **Systems:** Windows Desktop, Ubuntu Server
• **Libraries:** OpenCV, PyTorch, Halcon
• **IT Tools:** Git, Docker, Anaconda
• **Other Skills:** gdb/pdb, Keil (STM32), Altium Designer, Rhino, Dreamweaver
• **Books:** Computer Vision: Algorithms and Applications, Design Patterns: Elements of Reusable Object-Oriented Software
• **Patents:** CN202311136494.1; CN202310973415.6; CN202310287581.0; CN202310051665.4; CN202211602503.7; CN202211474648.3; CN202211357774.0; CN202210994137.8; CN202230264395.1; CN201911330078.9; CN201110120104; CN201110308514.X